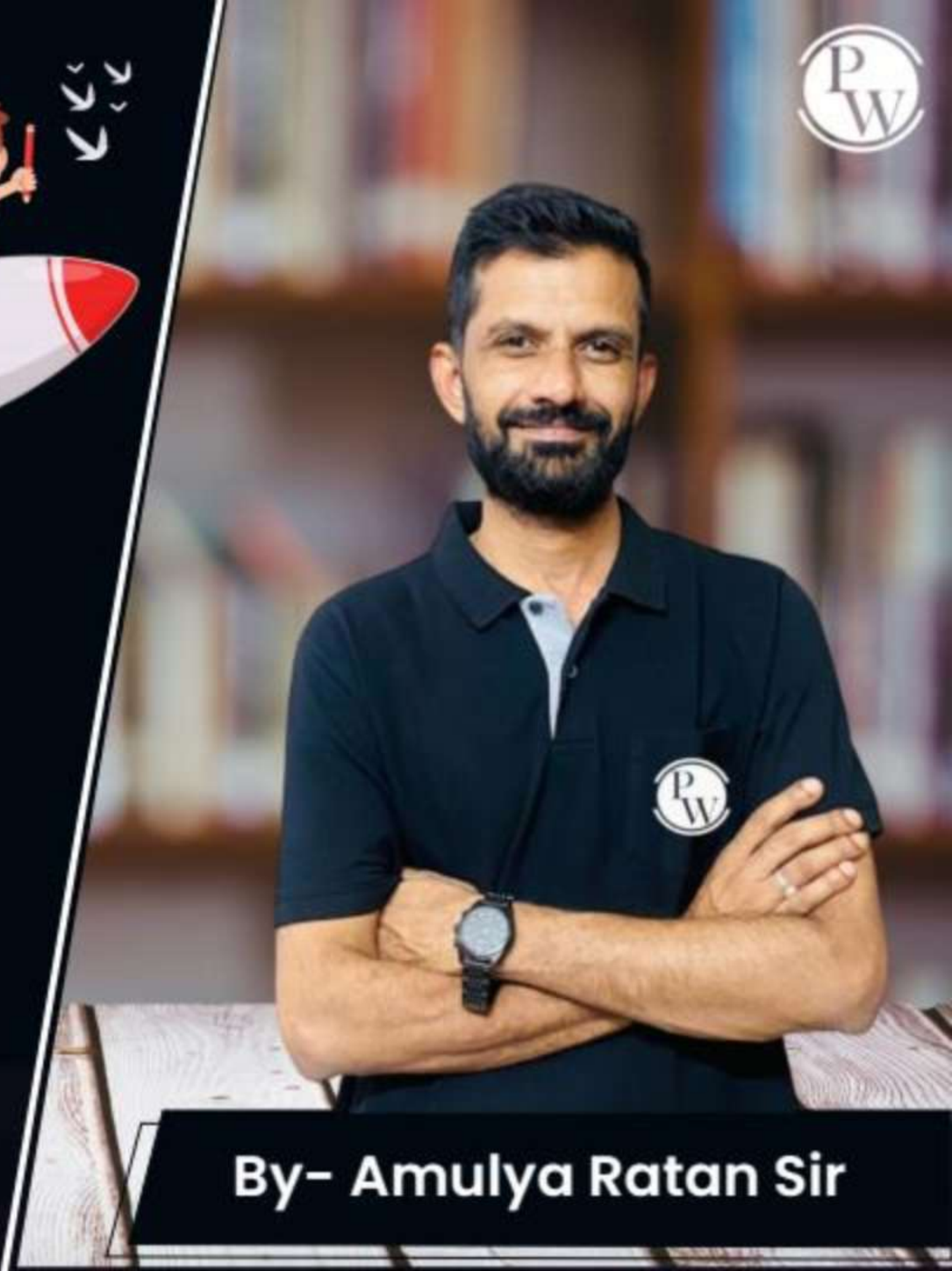
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 08



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Mixtures Alligations & Basics of Numbers

Topics to be Covered



Topic-1 Understanding Number System

Topic-2 Divisibility Rules

Topic-3 Questionnaire

[MCQ]

#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

Assignment

A 3:8

B 5:3

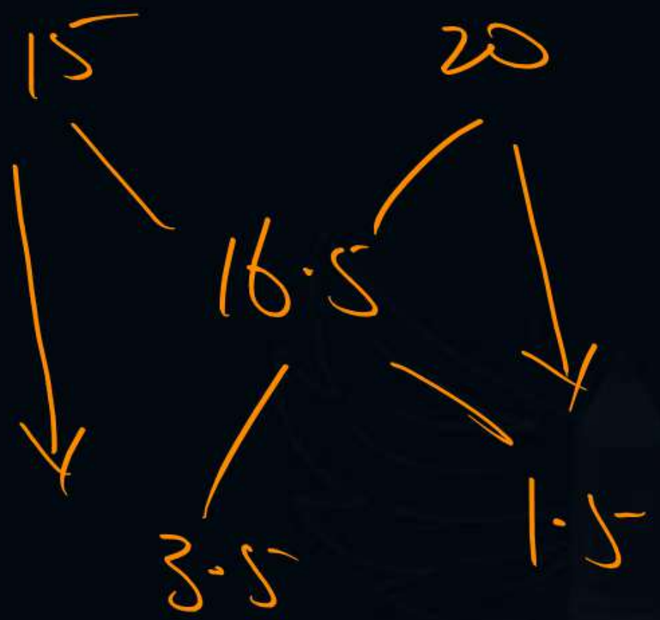
C 7:3

D 4:5

3.5 : 1.5

= 35 : 15

7 : 3



[MCQ]

#Q. In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

Assignment

A 1:5

B 3:5

C 2:3

D 3:2



S.P = 36

$\frac{36}{C.P} = 1.2$

$\frac{36}{C.P} = \frac{30}{12}$

C.P

1

2 : 10

1 : 5

Number System

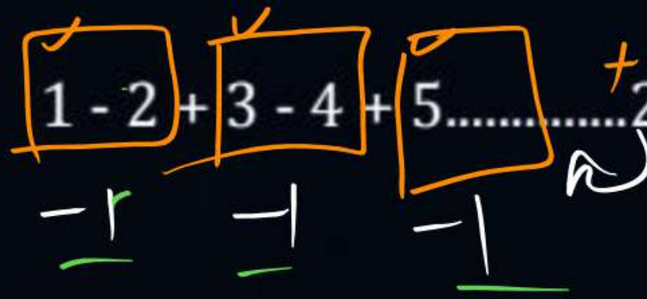
Basics

Twin ✓

Co-prime ✓

[MCQ]

#Q. Find the answer of given expression: $1 - 2 + 3 - 4 + 5 \ldots\ldots\ldots\ldots 2023$.



2023

-1011

1012

2022

[MCQ]

#Q. What would the unit digit in the answer of given expression:

$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2024!$ = Ans ?

3

1! = 1

2! = 2

3! = 6

4! = 24

5! = 120

9

120 × 6 = 720

720 × 7 = 5040

1

[MCQ]

#Q. What would the remainder when the answer of given expression is divided by 3?

$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2024!$

Multiple of 3

(A) 1

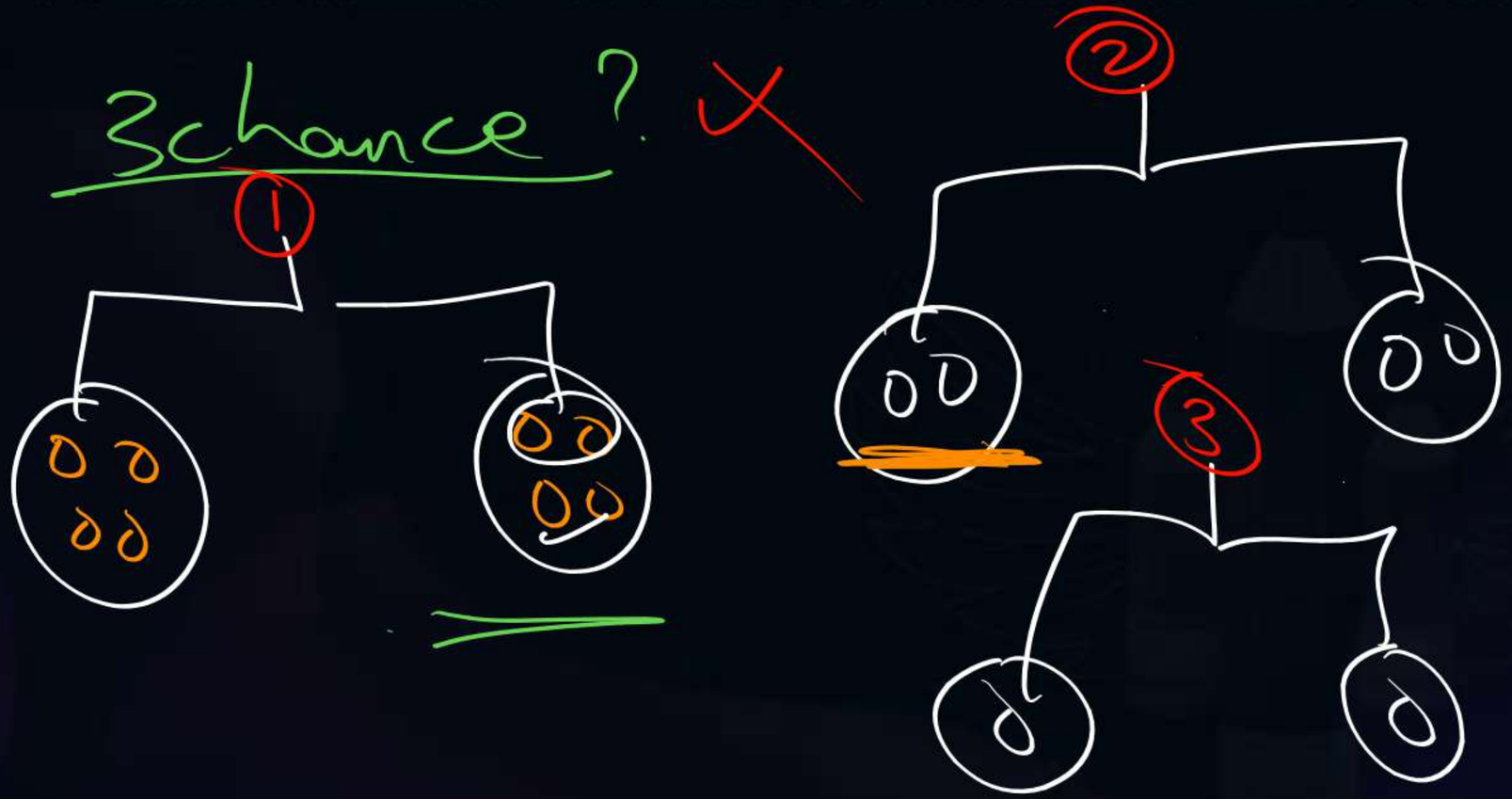
(B) 2

(C) 0

(D) Can't be determined

[MCQ]

#Q. There are 8 balls of which one is defective.Given that the defective ball is of less weight and remaining are of equal weights. What are the minimum number of chances a common balance is to be used to find the defective one?



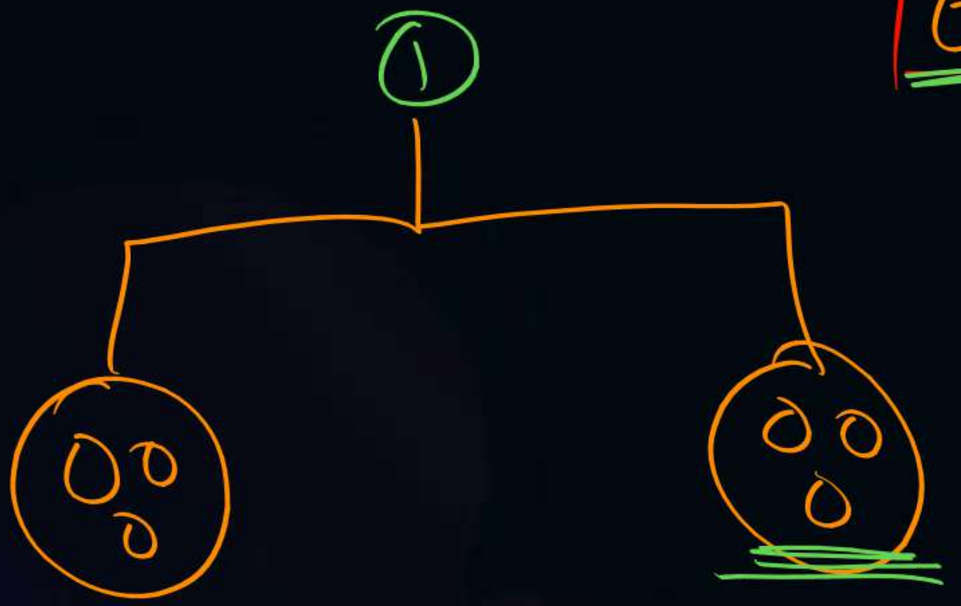
8balls

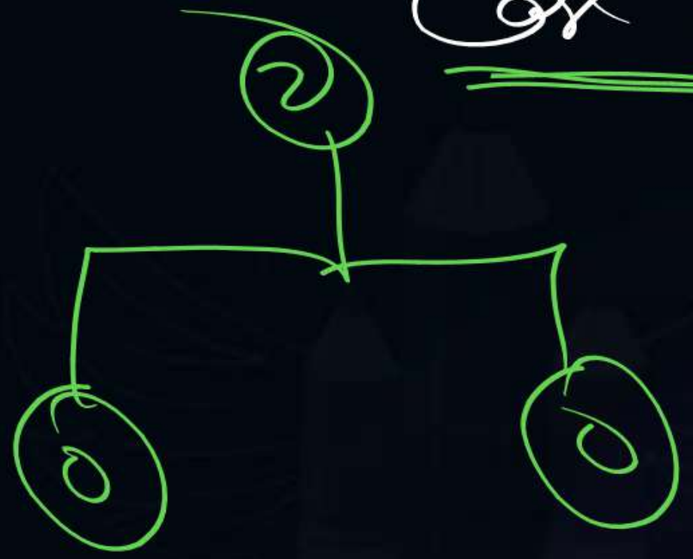
6ball

2ball

Case 1 =

Case 2 ≠





[MCQ]

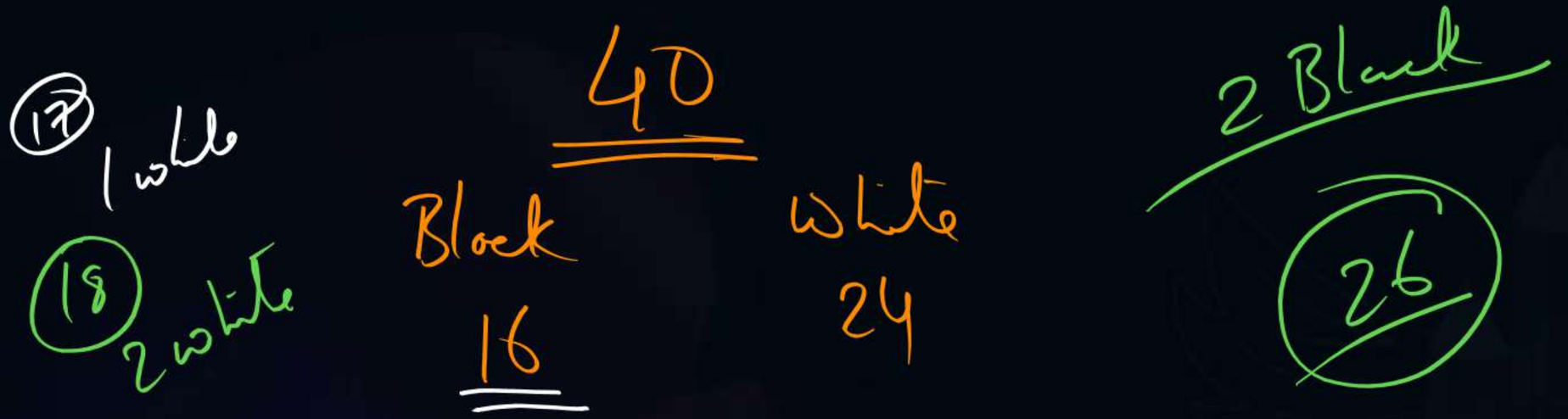
#Q. A bag consist of 48 red colour balls, 16 green colour balls, 12 yellow colour balls, 11 black colour balls and 11 white colour balls. How many minimum number of balls are to be taken out from the bag randomly so that we get atleast two balls of same colour?

blindfolded

"6 balls"

[MCQ]

#Q. A shelf consist of 40 socks. 40% of these are black and remaining are white. How many minimum number of socks are to taken out from the shelf randomly (blindly), so that we get atleast two white socks?



[MCQ]

#Q. A box consist of 40 pairs of shoes of equal size. 40% of these are black and remaining are white. How many minimum number of shoes are to taken out from the box randomly, so that we get atleast a pair of black shoes?



[MCQ]

#Q. How many zeroes ~~would be~~ at the end in the answer of 50! ?

$25 = 5 \times 5$

$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \ldots 50$

A 12

B 15

C 23

D 24

$\frac{50}{5} = 10$

$\frac{10}{5} = 2$

$\frac{2}{5} = 0$

12

$1 - 10 \Rightarrow 2$

$11 - 20 \Rightarrow 2$

$21 - 30 \Rightarrow 3$

$31 - 40 \Rightarrow 2$

$41 - 50 \Rightarrow 3$

25×2

12

[MCQ]

#Q. If 50! can be denoted maximum 7^x, then the value of x is?

A 9

B 8

C 7

D None of these

$$\frac{50}{7} = 7$$

$$\frac{7}{7} = 1$$

$$\frac{1}{7} = 0$$

$$8$$

[MCQ]

#Q. Find the number of zeroes in following multiplication:

$5 \times 10 \times 15 \times 20 \times 25 \ldots\ldots \times 50$?

Assignment

A 48

B 12

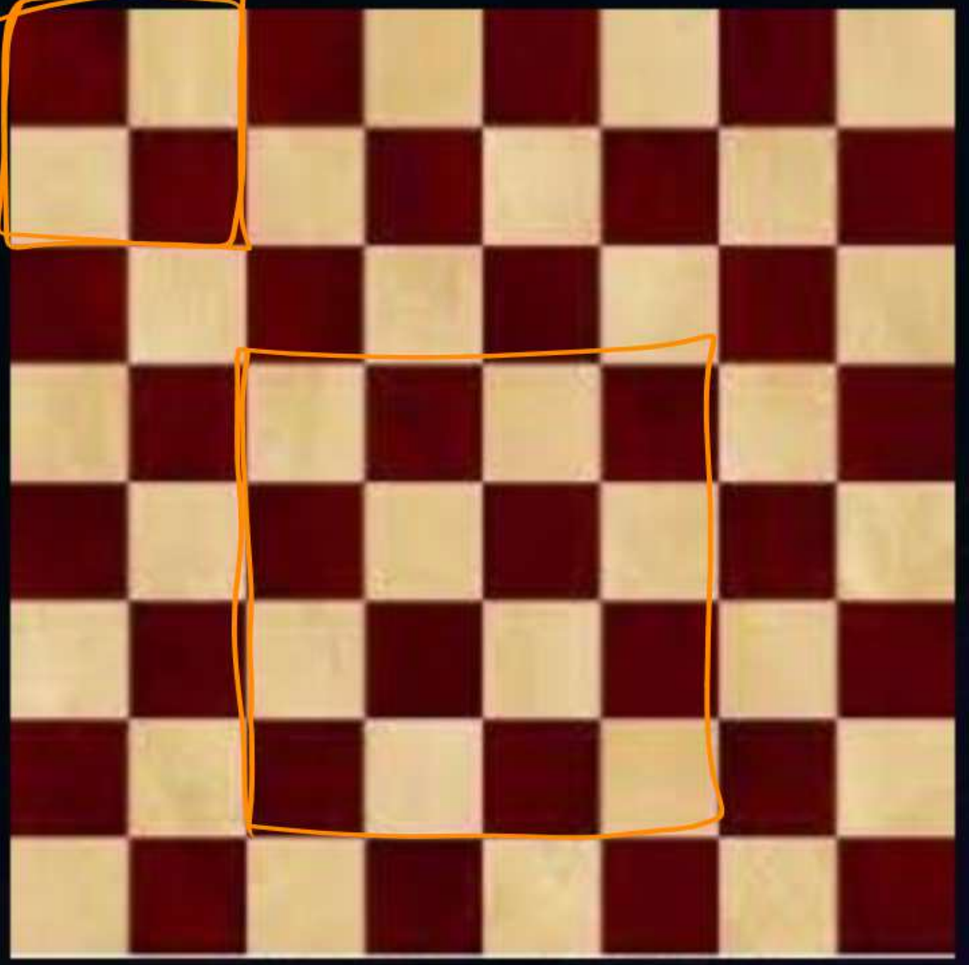
C 24

D 8

[MCQ]

#Q. How many squares are there in a chessboard?

Assignment





2 mins Summary

Topic Number System

THANK - YOU